Calculus 1 Homework 1 1.11.26

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1. $A=\{x\in\mathbb{R}: \frac{x-3}{x+1}\geq -1\} \longrightarrow \frac{x-3}{x+1}\geq -1 \longrightarrow \frac{x-3}{x+1}+1\geq 0 \longrightarrow \frac{2x-2}{x+1}\geq 0 \longrightarrow A=\{x\in\mathbb{R}: x<-1 \cup 1\leq x\}$

$B=\{x\in\mathbb{R}: |x+2|-|x|>1\}$

① $x\geq 0$: $x+2-x>1$ true $\forall x\leq 0$

② $-2\leq x<0$: $x+2+x>1 \rightarrow -\frac{1}{2}<x$

③ $x<-2$: $-x-2+x>1$ false $\forall 2<x$

$B=\{x\in\mathbb{R}: -\frac{1}{2}<x\}$

$C=\{x\in\mathbb{R}: x^2\leq 3\} \rightarrow C=\{x\in\mathbb{R}: -\sqrt{3}\leq x\leq\sqrt{3}\}$

$D=\{x\in\mathbb{R}: -8<x<-1\}$

$E=\{x\in\mathbb{R}: \sqrt{-x}>0\} \rightarrow E=\{x\in\mathbb{R}: x<0\}$

$C\cap\mathbb{Z}=\{0,\pm 1\}$

$A^c\cup D=\{x\in\mathbb{R}: -1\leq x<1\}\cup\{x\in\mathbb{R}: -8<x<-1\}=\{x\in\mathbb{R}: -8<x<1\}$

$B\cap E=\{x\in\mathbb{R}: -\frac{1}{2}<x\}\cap\{x\in\mathbb{R}: x<0\}=\{x\in\mathbb{R}: -\frac{1}{2}<x<0\}$

$C\setminus B=\{x\in\mathbb{R}: -\sqrt{3}\leq x\leq\sqrt{3}\}\setminus\{x\in\mathbb{R}: -\frac{1}{2}<x\}=\{x\in\mathbb{R}: -\sqrt{3}\leq x\leq -\frac{1}{2}\}$

2. c) $A\subseteq B\Rightarrow A\setminus C\subseteq B\setminus C$

Let $x\in A\setminus C \rightarrow x\in A\wedge x\notin C$

Since $A\subseteq B\Rightarrow x\in B$

Since $x\in B\wedge x\notin C\Rightarrow x\in B\setminus C\Rightarrow A\setminus C\in B\setminus C$ True

d) $A\setminus(B\setminus C)=(A\setminus B)\setminus C$ False

$A=\{1,2,3\}\ B=\{1,2\}\ C=\{2\}$

$A\setminus(B\setminus C)=\{1,2,3\}\setminus\{1\}=\{2,3\}$

$(A\setminus B)\setminus C=3$

2. c) $A \cup B = A \cap B \Rightarrow A = B$ True

* Let $x \in A \rightarrow x \in A \cup B$

$A \cup B = A \cap B \Rightarrow x \in A \cap B \Rightarrow x \in B \Rightarrow A \subseteq B$

* Let $y \in B \rightarrow y \in A \cup B$

$A \cup B = A \cap B \Rightarrow y \in A \cap B \Rightarrow y \in A \Rightarrow B \subseteq A$

$A = B$

3. $A = \{x \in \mathbb{R} : 0 < x < 2\}$ $B = \{y \in \mathbb{R} : 1 < y < 4\}$

a) $\forall x \in A \; \exists y \in B : y = x^2 + 1$

$0 < x < 2 \Rightarrow 0 < x^2 < 4 \Rightarrow 1 < x^2 + 1 < 5$ False

b) $\forall y \in B \; \exists x \in A : x^2 = y - 1$

$1 < y < 4 \Rightarrow 0 < y - 1 < 3 \Rightarrow 0 < \sqrt{y-1} < \sqrt{3} < 2$ True

d) $\forall x \in A, \forall y \in B : x < y \Rightarrow \frac{x}{y} < \frac{2}{3}$

Let $x = 1.8$, $y = 2$ $\frac{x}{y} = \frac{9}{10} > \frac{2}{3}$ False

4. a) $1^3 + 2^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$

$n = 1$: $1^3 = \frac{1^2 2^2}{4} = 1$ ✓

given for n is true:

$1^3 + 2^3 + (n+1)^3 = (1^3 + 2^3 + \dots + n^3) + (n+1)^3 = \frac{n^2(n+1)^2}{4} + (n+1)^3 = \frac{n^2(n+1)^2 + 4(n+1)(n+1)^2}{4} = \frac{(n+1)^2(n^2+4n+4)}{4} = \frac{(n+1)^2(n+2)^2}{4}$ ✓

b) $\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdot \dots \cdot \frac{2n-1}{2n} \le \frac{1}{\sqrt{2n+1}}$

$n = 1$: $\frac{1}{2} \le \frac{1}{\sqrt{3}}$ ✓

given for n is true:

$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdot \dots \cdot \frac{2n-1}{2n} \cdot \frac{2(n+1)-1}{2(n+1)} = \left(\frac{1}{2} \cdot \frac{3}{4} \cdot \dots \cdot \frac{2n-1}{2n}\right) \frac{2n+1}{2n+2} = \frac{1}{\sqrt{2n+1}} \cdot \frac{2n+1}{2n+2} = \frac{\sqrt{2n+1}}{2n+2} \le \frac{1}{\sqrt{2n+3}}$ ✓

$\left(\frac{\sqrt{2n+1}}{2n+2}\right)^2 \le \left(\frac{1}{\sqrt{2n+3}}\right)^2$

$\frac{2n+1}{4n^2+8n+4} \le \frac{1}{2n+3}$ / all are > 0

$4n^2 + 8n + 3 \le 4n^2 + 8n + 4$

$3 \le 4$ ✓

4. c) $\tan(nx) \geq n \cdot \tan x$ for $0 < x < \frac{\pi}{2n}$

$n = 1$: $\tan x \geq \tan x$ ✓

given for n is true:

$$\tan((n+1)x) = \tan(nx+x) = \frac{\tan(nx)+\tan x}{1-\tan(nx)\tan x} \geq \frac{(n+1)\tan x}{1-n\tan^2 x} \geq \frac{(n+1)\tan x}{1} = (n+1)\tan x$$

5. Rational number: $\frac{m}{n}$, $n \neq 0$ — m, n - have no common divider

Assume $\sqrt{2}$ is rational: $\sqrt{2} = \frac{m}{n}$

$2 = \frac{m^2}{n^2}$ $m^2 = 2n^2 \rightarrow$ m^2 is even $\rightarrow$ m is even

m can be represented as $2k$

$m = 2k \rightarrow m^2 = 4k^2 = 2n^2$

$2k^2 = n^2 \rightarrow n^2$ is even $\rightarrow n$ is even

m, n have a common divider: 2. Our assumption is wrong